

# 1D-Silver-Price-Forecasting-Agent - Final Report

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GitHub: [github.com/iemunkaloo25-hub/1D-Silver-Price-Forecasting-Agent](https://github.com/iemunkaloo25-hub/1D-Silver-Price-Forecasting-Agent)

## 1. INTRODUCTION & PROBLEM STATEMENT

Silver commodity prices are highly volatile and influenced by global economic factors. The objective is to build an accurate forecasting agent to predict future silver prices for trading decisions.

## 2. DATASET & EDA

- Dataset: Silver Price Historical Data
- Trend: Upward with high volatility
- Feature Engineering: Lag\_1, Lag\_7, MA\_7, MA\_30, RSI
- Split: 80% Training, 20% Testing

## 3. METHODOLOGY

| Model        | RMSE   | Status         |
|--------------|--------|----------------|
| ARIMA        | 0.89   | Baseline       |
| Prophet      | 0.76   | Seasonal Model |
| LSTM         | 0.65   | Deep Learning  |
| XGBoost      | 0.1765 | BEST MODEL     |
| Ensemble Avg | 1.0461 | Failed         |

## 4. RESULTS & ANALYSIS

- Winner: XGBoost with RMSE 0.1765 (lowest error)
- LSTM second best but overfitted on small data
- Ensemble failed because weak ARIMA dragged average down
- Simple Best > Complex Average
- SHAP: Lag\_1 and MA\_7 most important

## 5. FINAL DASHBOARD

Dashboard created (Day\_10\_Dashboard.png) showing Actual vs Predicted and 30-day forecast. XGBoost curve closely follows actual.

## 6. CONCLUSION

- Final Model: XGBoost (RMSE 0.1765)
- Ready for deployment
- Future: Add news sentiment, gold price, USD index

## 7. KEY LEARNINGS

- XGBoost > LSTM for tabular time-series
- Feature engineering > Model complexity
- Proper RMSE evaluation is critical

**Project Status: 100% Complete | Thank you ZeTheta!**